

Project Goodnight Report Update

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Project overview

Project Goodnight examines how sleep quality varies across urban areas within cities in China. Sleep quality is the primary outcome. In our regression models, perceived stress is the main predictor, and we study how district/county-level environmental and infrastructure measures are associated with sleep quality and whether they moderate the relationship between stress and sleep (i.e., stress $\times$ environment interactions). The goal is descriptive and correlational inference rather than causal claims, and we include observed covariates to adjust for basic differences across districts/counties.

Data status

1) PBICR-2023 Survey (confidential)¹

Status: Collected; ongoing cleaning and aggregation

Included in original proposal: Yes

Notes: Includes measures of sleep quality, stress, location and demographic variables. Participants have address on the district/county level.

2) Google Earth Engine (API)

Link: <https://code.earthengine.google.com/>

Status: Implemented for four cities, including Beijing, Suzhou, Chengdu and Shenzhen.

Included in original proposal: Yes

Notes: Satellite-derived measures include nighttime light and vegetation (NDVI). The pipeline has been validated on one city and is being scaled to the remaining cities.

3) Gaode (Amap) Places API

Link: <https://lbs.amap.com/>

Status: Implemented for four cities, including Beijing, Suzhou, Chengdu and Shenzhen.

Included in original proposal: Yes

Notes: Used to count hospitals, clinics, parks, gyms, and related facilities. API limits and POI category definitions are still being tested to ensure cross-city consistency.

Analysis plan update

We will not collect data from the 58 Tongcheng website anymore. Everything else remains the same as in the proposal.

¹ Study protocol: A national cross-sectional study on psychology and behavior investigation of Chinese residents in 2023, PBICR, <https://onlinelibrary.wiley.com/doi/full/10.1002/hcs2.125>

Planned outputs:

Choropleth maps showing average sleep quality by district/county (our main outcome).

Choropleth maps showing the spatial distribution of key predictors, including perceived stress and district/county-level environmental and infrastructure measures (e.g., green space, POI counts).

Bar charts and tables comparing these measures across districts/counties and across cities.

Descriptive summaries (summary statistics and basic correlations/scatterplots) to document overall patterns.

Regression results reported in tables and visualized with coefficient plots (with confidence intervals) and predicted values/marginal effects plots for the main predictors (and interactions if included).

Timeline / next steps

Extend environmental data collection to cities beyond the four currently implemented in the repository if the model does not go well.

Continue cleaning and harmonizing geographic identifiers across all data sources.

Merge all sources into a single district/county-level dataset. Match the survey and collected data.

Produce initial maps and summary tables for exploratory reporting.

Run data analysis, like regression model .

Team responsibilities and plans**Shoshana Abikzer**

The 5th week: 1. Collects environmental data using Google Earth Engine

2. Processes nighttime light and NDVI measures. Aggregates environmental data to the district/county level

3. Produces initial summary tables and visualizations

The 6th week: Write the paper

The 7th week: prepare the video and the presentation

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The 5th week: 1. Collects point-of-interest data using the Gaode Places API.

2. Constructs access measures for health and recreational facilities

The 6th week: 1. Match the data from the survey and data collected by API. 3. Do the data cleaning of the survey and data analysis.

The 7th week: prepare the video and the presentation